

$$= \frac{I}{4\pi P} 2 \sin \alpha_2 = \frac{I}{2\pi P} \sin (180/n)$$

Put

$$P = R \cos \left(\frac{180}{n} \right)$$

$$\therefore H = \frac{I}{2\pi R} \tan \left(\frac{180}{n} \right)$$

$$\text{For } n \text{ sides, } \bar{H} = \frac{nI}{2\pi R} \tan \left(\frac{180}{n} \right)$$

$$\therefore \bar{B} = \mu \bar{H} = \frac{\mu n I}{2\pi R} \tan \left(\frac{180}{n} \right)$$

$$\text{as } n \rightarrow \infty, \frac{180}{n} \rightarrow 0, \tan \left(\frac{180}{n} \right) \rightarrow \left(\frac{180}{n} \right) = \frac{\pi}{n}$$

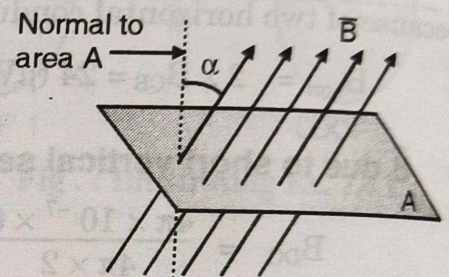
$$\therefore B = \frac{\mu n I}{2\pi R} \left(\frac{\pi}{n} \right) = \frac{\mu I}{2R} \text{ (T) (into the page)}$$

...A

7.4.3 Magnetic Flux (ψ_m) :

The magnetic flux crossing any surface is found by multiplying magnetic flux density with area of the surface, provided that B is normal to the surface. If B makes some angle with surface not equal to 90° , then

$$\psi_m = \iint_S \bar{B} \cdot d\bar{s} \quad \dots(7.4.2)$$



Where, ψ_m = magnetic flux (Wb)

$\bar{B}$ = Magnetic flux density (Wb/m² or T)

Fig. 7.4.1 : Flux density not in normal to the area

7.4.4 Solved Examples on Magnetic Flux :

Important Formula

$$\psi_m = \iint_S \bar{B} \cdot d\bar{s}$$

Example 7.4.4 : A radial field

$$\bar{H} = \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r \text{ (A/m)}$$

exists in free space. Find the magnetic flux crossing the surface defined by $0 \leq \phi \leq \pi/4$ and $0 \leq z \leq 1$ m.

Solution : Magnetic flux density is obtained from magnetic field intensity using the relation,

$$\bar{B} = \mu_0 \bar{H} = 4\pi \times 10^{-7} \times \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r$$

$$= \frac{3.0}{r} \cos \phi \bar{a}_r \text{ (T)}$$

Magnetic flux crossing the given surface is obtained by $\phi = \iint \bar{B} \cdot d\bar{s}$

For the given surface ϕ and z are varying therefore the third co-ordinate is constant i.e. $r = \text{constant}$. Then,

$$d\bar{s} = r d\phi dz \bar{a}_r$$

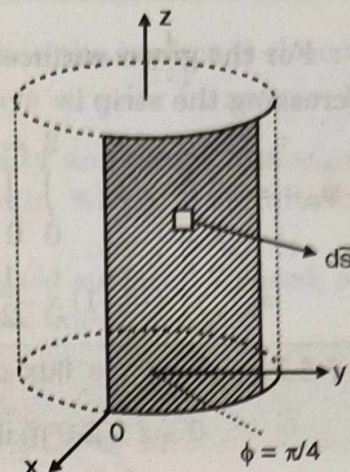


Fig. : Illustrating Example 7.4.4

Thus,

$$\begin{aligned} \psi_m &= \int_0^1 \int_0^{\pi/4} \left(\frac{3.0}{r} \cos \phi \bar{a}_r \right) \cdot (r d\phi dz \bar{a}_r) = 3.0 \times 1 \times \int_0^{\pi/4} \cos \phi d\phi \\ &= 3.0 [\sin \phi]_0^{\pi/4} = 2.12 \text{ (Wb)} \end{aligned}$$

...Ans.

Example 7.4.5 : A radial field $\bar{H} = \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r$ (A/m) exits in

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free space. Find the magnetic flux ϕ crossing the surface defined by $-\pi/4 \leq \phi \leq \pi/4$, $0 \leq z \leq 1$ m.

Solution : Magnetic flux density is obtained from magnetic field intensity using the relation

$$\bar{B} = \mu_0 \bar{H} = 4\pi \times 10^{-7} \times \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r = \frac{3.00}{r} \cos \phi \bar{a}_r \text{ (T)}$$

Magnetic flux crossing the given surface is

$$\psi_m = \iint \bar{B} \cdot d\bar{s}$$

For the given surface ϕ and z are varying, therefore the third co-ordinate is constant i.e.

$r = \text{constant}$. This means that the differential area is $d\bar{s} = r d\phi dz \bar{a}_r$

$$\therefore \psi_m = \int_0^1 \int_{-\pi/4}^{\pi/4} \left(\frac{3.0}{r} \cos \phi \bar{a}_r \right) \cdot (r d\phi dz \bar{a}_r) = 3.0 \times 1 \times \int_{-\pi/4}^{\pi/4} \cos \phi d\phi$$

$$= 3.0 \times [\sin \phi]_{-\pi/4}^{\pi/4} = 4.24 \text{ (Wb).}$$

...Ans.

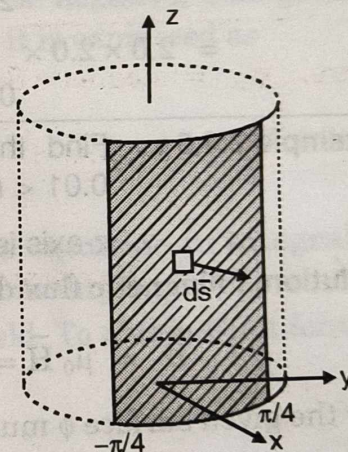


Fig. : Illustrating Example 7.4.5

Example 7.4.6 : Given that $\bar{B} = 2.5 \left(\sin \frac{\pi x}{2} \right) e^{-2y} \bar{a}_z$ Telsa, find the total magnetic flux crossing the strip $z = 0$, $y \geq 0$, $0 \leq x \leq 2$ m.

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This gives a flux crossing the open surface. What happens when the surface is closed ?

We know that the magnetic field intensity H is always circular in nature. Due to relation between $\vec{B}$ and $\vec{H}$ which is, $\vec{B} = \mu\vec{H}$, the flux density and hence flux ψ_m will be also circular in nature. For the standard source of magnetic, which is infinite current filament, the flux lines are as shown in Fig. 7.4.2.

In the presence of these magnetic flux lines when a closed surface is placed, one can very easily observe that number of incoming and outgoing flux lines are equal.

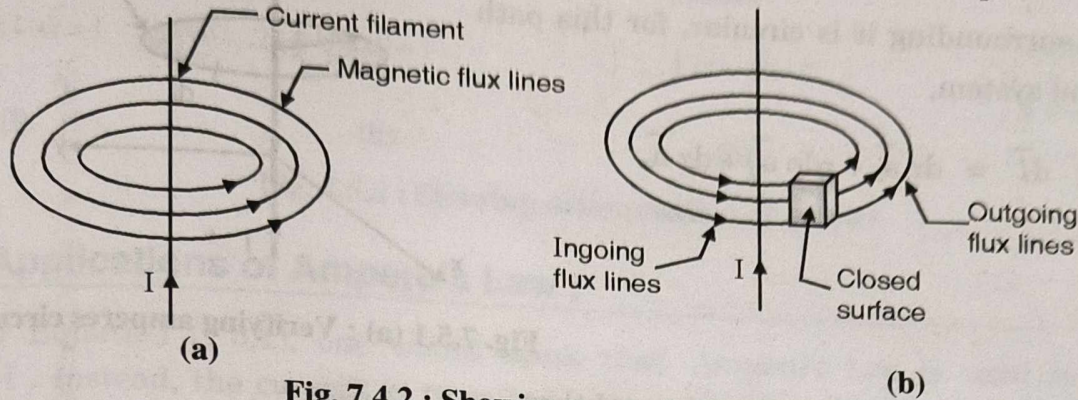


Fig. 7.4.2 : Showing magnetic flux lines

If incoming flux is considered positive then the outgoing flux will be negative. This gives **total flux crossing the closed surface as zero**. Mathematically it is expressed as

$$\oint_S \vec{B} \cdot d\vec{s} = 0 \quad \dots(7.4.3)$$

This is called as Gauss's law for magnetic, also called as **Maxwell's equation in integral form** derived from Gauss's law for static field. In the chapter on time varying field.

We shall see that this law is used as it is for time varying field. To obtain point form of this equation, apply divergence theorem to above equation.

$$\oint \vec{B} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{B}) dv = 0$$

i.e.

$$\boxed{\nabla \cdot \vec{B} = 0}$$

...(7.4.4)

This is called as **Maxwell's equation in point form** derived from Gauss's law for static field.

7.5 Ampere's Circuital Law or Ampere's Work Law :

Statement :

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The line integral of $\vec{H}$ around a single closed path is equal to the current enclosed by that path.

Mathematically,

$$\boxed{\oint \vec{H} \cdot d\vec{l} = I}$$

... (7.5.1)